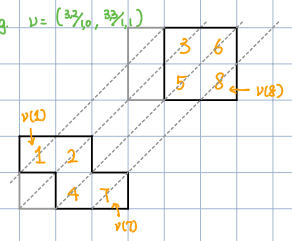
Let ν be a tuple of skew shapes. Denote $\nu(1), \nu(2), \dots, \nu(\ell)$ the boxes of ν in increasing reading order (see §2.2) and set $\nu(I) := \{\nu(i) : i \in I\}$ for any $I \subseteq [\ell]$.

e.g. $\nu = (\,^{32}/_{10}, \,^{32}/_{11})$

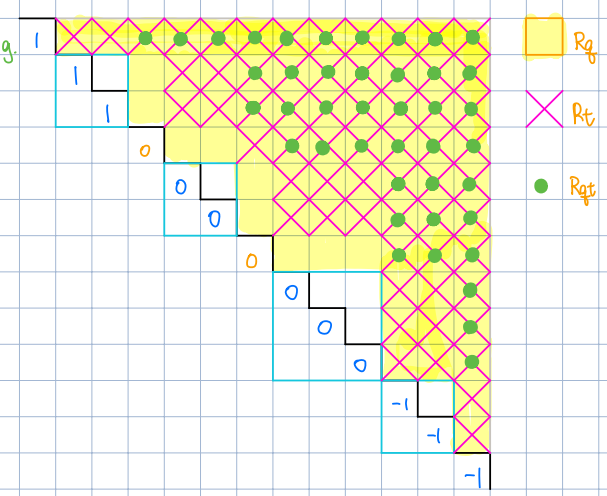


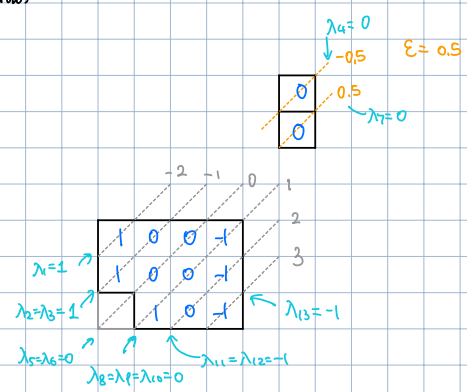
Def 7.1.2: The ($(1,0)$ case) LLT Catalanimal associated to a tuple of skew shapes ν with a total of ℓ boxes is the length ℓ Catalanimal $H_\nu = H(R_q, R_t, R_{qt}, \lambda)$ given by

- $R_q = \{\alpha_{ij} \in R_+ \mid \tilde{c}(\nu(i)) < \tilde{c}(\nu(j))\}$
- $R_t = \{\alpha_{ij} \in R_+ \mid \tilde{c}(\nu(i)) + 1 \leq \tilde{c}(\nu(j))\}$
- $R_{qt} = \{\alpha_{ij} \in R_+ \mid \tilde{c}(\nu(i)) + 1 < \tilde{c}(\nu(j))\}$
- $\lambda_i = \mathbb{1}(\text{diag}(\nu(i)) \text{ contains the } 1^{st} \text{ box in a row}) - \mathbb{1}(\text{diag}(\nu(i)) \text{ contains the last box in a row})$

where $\text{diag}(\alpha)$ = diagonal of ν containing α.

e.g.





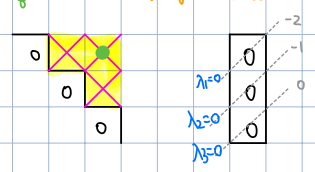
Remark: (i) $R_+ \supseteq R_q \supseteq R_t \supseteq R_{qt}$, and

- $R_+ \setminus R_q = \{\alpha_{ij} \in R_+ \mid \nu(i) \text{ and } \nu(j) \text{ are on the same diagonal}\}$ — hence must be in the same skew shape of ν
- $R_q \setminus R_t = \{\alpha_{ij} \in R_q \mid \nu(i) \text{ and } \nu(j) \text{ form an attacking pair}\}$
- $R_t \setminus R_{qt} = \{\alpha_{ij} \in R_t \mid \nu(i) \text{ and } \nu(j) \text{ are on adjacent diagonals}\}$ — if adjusted content differ by 1 (hence must be in the same shape)
- $R_{qt} = \{\text{all others}\}$

(ii) The data R_q, R_t, R_{qt} and λ are constant on diagonals of ν, in the sense that λ_i depends only on $\text{diag}(\nu(i))$, and whether or not α_{ij} belongs to R_q (likewise for R_t, R_{qt}) depends only on $\text{diag}(\nu(i))$ and $\text{diag}(\nu(j))$.

(iii) One way to obtain λ is to fill in 1 in the leftmost cell and -1 in the rightmost cell in each row and 0 the rest.
If the row has only 1 cell, then fill it with 0 (b/c $1+(-1)=0$)
The λ_i = sum of numbers in all the cells in $\text{diag}(\nu(i))$ (see example above).

e.g. $\nu = ((1,1,1))$, then $R_q = R_t = R_+ = \{\alpha_{12}, \alpha_{13}, \alpha_{23}\}$ ($R_+ \setminus R_q = \phi$ b/c each diagonal has only 1 cell. $R_q \setminus R_t = \phi$ b/c there is only one shape, i.e. no attacking cells)
$R_{qt} = \{\alpha_{13}\}$ (b/c $R_t \setminus R_{qt} = \{\alpha_{12}, \alpha_{23}\}$ b/c $\nu(1), \nu(2)$ and $\nu(2), \nu(3)$ are on adjacent diagonals)



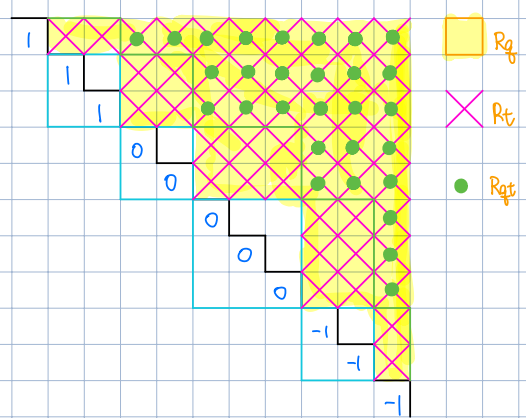
$$H_\nu = \sum_{w \in S_3} w\left(\frac{1 - \frac{qt}{b}\frac{z_3}{z_3}}{\prod_{1 \leq i < j \leq 3}\left(1 - \frac{z_i}{z_j}\right)\left(1 - q\frac{z_i}{bz_j}\right)\left(1 - t\frac{z_i}{z_j}\right)}\right)$$

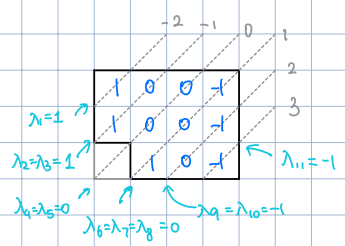
e.g. ν : single ribbon skew shape θ, then $R_g = R_t = R_+$ and $R_{gt} = [R_+, R_+] = \{\alpha_{ij} \in R_+ \mid j-i > 1\}$

e.g. ν: single skew shape, then $R_g = R_t = \{\alpha_{ij} \in R_+ \mid \alpha_{ij}$ is outside the diagonal blocks with block sizes = length of diag.s of shape $\theta\}$

$R_{gt} = \{\alpha_{ij} \in R_+ \mid \alpha_{ij}$ is outside the blocks on or "between" adjacent diagonal blocks$\}$

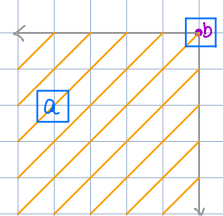
for example:





Def: Define a partial order $\prec$ on boxes in ν by setting $a \preceq b$ if boxes a, b belong to the same skew shape $\nu_{(i)}$ and a is weakly SW of b, i.e. $a \leq b$ in the usual product order on $\mathbb{N} \times \mathbb{N}$. Let $R_t^{\prec} = \{\alpha_{ij} \in R_t \mid \nu(i) \prec \nu(j)\}$

i.e.



every lattice point a in the shaded region is less than b

Prop 7.1.6: The weight λ in Def 7.1.2 has the following alternative descriptions:

- $\lambda_i = \mathbb{1}(\text{diag}(\nu(i))$ does not contain the last box in a row$) - \mathbb{1}(\text{diag}(\nu(i))$ does not contain the first box in a row$)$
- $\lambda = -\sum(R_t \setminus R_g) + \sum((R_t \setminus R_{gt}) \cap R_t^{\prec})$

Proof: • $\lambda_i = \mathbb{1}(\text{diag}(\nu(i))$ contains the 1st box in a row$) - \mathbb{1}(\text{diag}(\nu(i))$ contains the last box in a row$)$

$= 1 - \mathbb{1}(\text{diag}(\nu(i))$ does not contain the 1st box in a row$) - (1 - \mathbb{1}(\text{diag}(\nu(i))$ does not contain the last box in a row$))$

$= \mathbb{1}(\text{diag}(\nu(i))$ does not contain the last box in a row$) - \mathbb{1}(\text{diag}(\nu(i))$ does not contain the first box in a row$)$

• Let $\mu = -\sum(R_t \setminus R_g) + \sum((R_t \setminus R_{gt}) \cap R_t^{\prec})$.

Consider the contribution to μ_i from each of the sums.

Let $D = \text{diag}(\nu(i))$ and C, E be the (possibly empty) adjacent diagonals with adjusted contents $\tilde{c}(\nu(i)) - 1$, $\tilde{c}(\nu(i)) + 1$ respectively.

Set $\nu_{\geq i} := \{a \in \nu \mid a \succeq \nu(i)\}$, $\nu_{\prec i} = \{a \in \nu \mid a \prec \nu(i)\}$, we have

$$\mu_i = -\Big(\underbrace{|D \cap \nu_{\geq i}|}_{\alpha_{ij} = \varepsilon_i - \varepsilon_j} - \underbrace{|D \cap \nu_{\prec i}|}_{\alpha_{ji} = \varepsilon_j - \varepsilon_i}\Big) + \Big(-\underbrace{|C \cap \nu_{\prec i}|}_{\alpha_{ji} = \varepsilon_j - \varepsilon_i} + \underbrace{|E \cap \nu_{\geq i}|}_{\alpha_{ij} = \varepsilon_i - \varepsilon_j}\Big) = \underbrace{\big(|E \cap \nu_{\geq i}| - |D \cap \nu_{\geq i}|\big)}_{\substack{=0 \text{ if } D \text{ contains the last box of some row} \\ =1 \text{ if } D \text{ does not contain the last box of some row}}} - \big(|C \cap \nu_{\prec i}| - |D \cap \nu_{\prec i}|\big)$$

$\mathbb{1}(D$ does not contain the last box of a row$)$

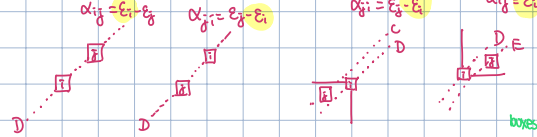
$=0$ if D contains the 1st box of a row

$=1$ if D does not contain the 1st box of a row

$\mathbb{1}(D$ does not contain the 1st box of any row$)$

boxes in D and C pair up vertically

boxes in D and E pair up vertically



and result follows.